AI-Powered Smart Shopping Cart System (Stage 1)

IoT Flow, Connections and Verified Component Models

Technical Report - Laptop, Camera and IoT Hardware (Without Cart)

22 October 2026

1. Introduction

1.1 Purpose of the Report

This report describes how the Stage 1 prototype of the AI-powered smart shopping cart works as an Internet of Things (IoT) system. It explains the flow of data from the sensors to the laptop, states how every component is connected, and names the specific, commercially available model of each component that works correctly in that connection. The Stage 1 blueprint drawing prepared for the project is included as Figure 1.

1.2 Scope of Stage 1

Stage 1 does not include the physical cart. It proves the sensing chain on a table: a USB camera and a YOLOv8 model identify the product, a load cell with an HX711 amplifier measures its weight, an ESP32 module reads the weight and drives the buzzer and light indicators, and the laptop combines both results and shows the detected items and the total. The same chain is later moved onto the cart, where the cloud, database and payment stages of the project are added.

2. Stage 1 System Blueprint

2.1 Description of the Blueprint

The blueprint (Figure 1, next page) contains six panels: the front view of the laptop and camera setup, the side view of the load cell platform, the top view of the component layout, the dimensions of each component, the wiring diagram, and the working flow. The drawing is shown in its original colours so that the colour coding used for the wires, boxes and indicators can be read directly. Table 1 explains what each colour in the drawing represents.

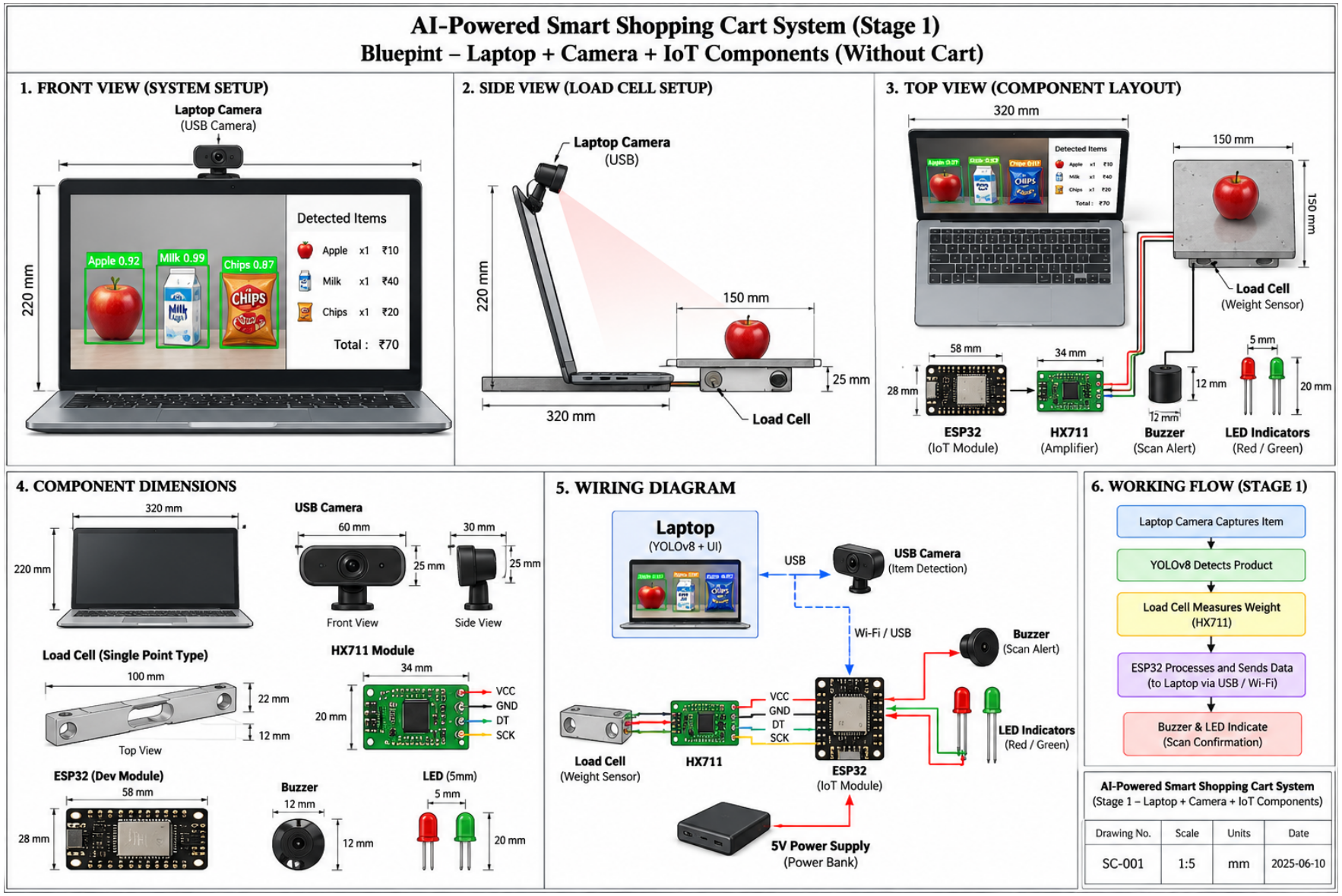


Figure 1. Stage 1 blueprint of the smart shopping cart: laptop, camera and IoT components (original colours).

2.2 Colour Legend of the Blueprint

The colours below are those used in the attached drawing. The report text itself is black and white; only the drawing is in colour.

Colour in drawing	Where it appears	Meaning
Light blue box	Working flow, step 1; laptop frame in the wiring diagram	Laptop and camera capture stage; host computer running YOLOv8 and the user interface
Green box	Working flow, step 2; bounding boxes on the screen	YOLOv8 product detection; each green rectangle is a detected item with its confidence score
Yellow box	Working flow, step 3; HX711 SCK line	Weight measurement by the load cell and HX711; yellow wire carries the clock (SCK) signal
Purple box	Working flow, step 4	ESP32 processing and sending of data to the laptop
Red / pink box	Working flow, step 5	Buzzer and LED indication of scan confirmation
Red wire or arrow	HX711 VCC line; supply to buzzer and LED; power bank arrow	Positive supply (VCC / 5 V) and the red LED signal path
Black wire	HX711 GND line; load cell wiring	Ground (0 V) common to all modules
Blue wire (HX711 side)	HX711 DT line	Digital weight data from HX711 to ESP32
Green wire	Green LED path; load cell signal wire	Green LED signal path; load cell signal output
Blue solid / dashed arrow	Laptop to camera and laptop to ESP32	Solid: USB cable. Dashed: Wi-Fi or USB data link
Red and green LED symbols	Components 4 and 5	Red: item rejected or error. Green: item confirmed

Table 1. Colour legend of the Stage 1 blueprint.

3. IoT Working Flow

3.1 End-to-End Flow

The Stage 1 system runs the following sequence each time an item is placed on the weighing platform.

Step 1 - Image capture. The USB camera, connected to the laptop, streams video frames to the laptop.

Step 2 - Product detection. The YOLOv8 model on the laptop detects the product in the frame and returns a class name with a confidence score.

Step 3 - Weight measurement. The load cell produces a very small differential voltage that changes with the load. The HX711 amplifies this signal and converts it into a 24-bit digital value.

Step 4 - Processing in the ESP32. The ESP32 reads the HX711 through its two-wire interface (DT and SCK), removes the empty-platform offset, applies the calibration factor and obtains the weight in grams. It then decides whether the weight is stable and whether it has increased or decreased.

Step 5 - Data transfer. The ESP32 sends the weight value to the laptop over USB serial or over Wi-Fi.

Step 6 - Fusion on the laptop. The laptop compares the detected product with the measured weight. The item is accepted only if both agree, and the item list and total are updated on the screen.

Step 7 - Confirmation. The laptop sends the result back, and the ESP32 sounds the buzzer and lights the green LED for an accepted item or the red LED for a rejected item.

3.2 Block Diagram of the Data Flow

Figure 2 shows the signal path between the components. The load cell side is analog and the remaining links are digital.

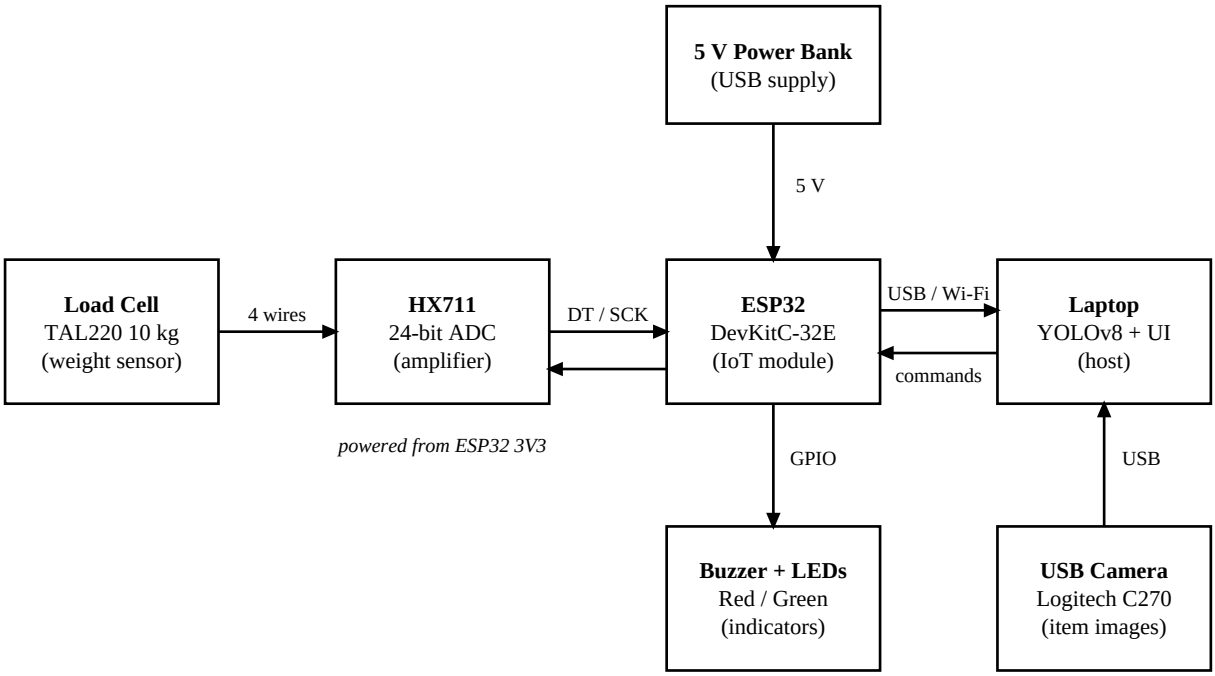


Figure 2. Block diagram of the Stage 1 IoT data flow (grayscale).

3.3 Sensor Fusion Rule

Neither sensor is trusted alone. The camera may confuse similar packages, and the load cell can be disturbed by vibration. Table 2 gives the decisions taken by the laptop.

Camera result	Weight result	Decision
Item found, confident	Weight rises as expected	Add item and update total
Item found, confident	Weight falls as expected	Remove item (put-back)
Item found	Weight does not match	Do not bill; red LED; flag for review
Not sure	Weight changes	Do not bill; ask the user to show the item again
Item seen	No weight change	Ignore; item was not placed on the platform

Table 2. Sensor fusion decisions.

4. Components and Verified Model Names

4.1 Component Selection

Table 3 lists the model recommended for each part of the Stage 1 blueprint. The specifications were checked against manufacturer or distributor documentation [1]-[6]. These models are chosen because their voltage levels, interfaces and dimensions are compatible with each other, which is what makes the connections in Section 5 work without damage to the boards.

No.	Component	Recommended model	Key specifications	Role in system
1	Host computer	Any laptop with a USB 2.0 or newer port, running Python 3	Runs Ultralytics YOLOv8, OpenCV and the user interface	Detection, fusion, display
2	USB camera	Logitech C270 HD Webcam (part no. 960-000694)	1280 x 720 at 30 fps; USB 2.0; fixed focus; clip mount; about 55 to 60 degrees diagonal field of view [6]. Logitech C920 is a 1080p alternative.	Item image capture
3	IoT module	Espressif ESP32-DevKitC-32E (ESP32-WROOM-32 E, 4 MB flash)	54.4 x 27.9 mm; micro-USB with USB-UART bridge; Wi-Fi and Bluetooth; 3.3 V logic on GPIO pins; 5 V supply through USB [3][4]	Reads HX711, drives indicators, talks to laptop
4	Amplifier / ADC	HX711 24-bit load cell amplifier module	Supply 2.6 to 5.5 V; channel A gain 128 or 64, channel B gain 32; 10 or 80 samples per second; current below 1.5 mA; two-wire DT and SCK interface [1][2]	Amplifies and digitises load cell signal

No.	Component	Recommended model	Key specifications	Role in system
5	Load cell	TAL220 10 kg straight bar load cell (SparkFun SEN-13329)	Aluminium alloy; 80 x 12.7 x 12.7 mm; IP66; two M4 and two M5 mounting holes; four wires: red E+, black E-, green O+, white O- [5]	Weight sensing under the 150 x 150 mm platform
6	Buzzer	12 mm active buzzer, 3.3 V to 5 V rating	Sounds on its own when powered; no tone generation needed	Scan alert
7	Indicators	5 mm LED, red and green, with 220 Ohm series resistors	About 6 mA at 3.3 V with 220 Ohm	Accepted / rejected indication
8	Buzzer driver (supporting)	NPN transistor BC547 or 2N2222 with 1 kOhm base resistor	Keeps buzzer current off the ESP32 pin	Switches buzzer
9	Power	5 V power bank with USB output (at least 1 A)	Powers ESP32 for standalone or Wi-Fi operation	Supply

Table 3. Recommended components for Stage 1 with model names.

4.2 Corrections to the Blueprint

While checking the blueprint against the component documentation, the following points were found that should be corrected before the circuit is built. They are listed in Table 4.

Item in blueprint	Issue	Correction
Load cell labelled single point type, 100 x 22 x 12 mm	The drawn size and shape do not correspond to the common low-cost bar cell used with HX711 modules.	Use the TAL220 10 kg straight bar cell, 80 x 12.7 x 12.7 mm. For lighter items a 1 kg to 5 kg bar cell of the same style gives finer resolution.
LEDs wired directly to ESP32	No current-limiting resistor is shown.	Add a 220 Ohm resistor in series with each LED.
Buzzer wired directly to ESP32	Buzzer current may exceed what a GPIO pin should supply.	Use an active buzzer through an NPN transistor with a base resistor.
HX711 supply not specified	A 5 V supply would raise the DT output above the 3.3 V limit of the ESP32 inputs.	Power the HX711 from the 3V3 pin of the ESP32. The ESP32 GPIO pins are not 5 V tolerant [4].
Camera labelled Laptop Camera (USB Camera)	Label is ambiguous; the drawn camera is an external clip-on USB camera.	Label it as USB Camera, Logitech C270.
ESP32 shown as generic module	No model named; pin availability depends on module.	Specify ESP32-DevKitC-32E (WROOM). GPIO 6 to 11 are reserved for flash and must not be used [4].
Title text	The word Blueprint is misspelled in the drawing title.	Correct to Blueprint.

Table 4. Corrections recommended for the Stage 1 blueprint.

5. Connections

5.1 Load Cell to HX711

The load cell is a Wheatstone bridge with four wires. Two wires supply the bridge and two return the signal. The colours in Table 5 apply to the TAL220 [5]; other makers may use different colours, so the datasheet must be checked.

Load cell wire	Function	HX711 terminal
Red	Excitation positive (E+)	E+
Black	Excitation negative (E-)	E-
Green	Signal positive (O+)	A+
White	Signal negative (O-)	A-

Table 5. Load cell to HX711 wiring.

If the weight value decreases when a load is added, swap the green and white wires.

5.2 HX711 to ESP32

The HX711 uses a two-wire serial interface, so any two free GPIO pins can be used. GPIO 16 and 17 are free on the WROOM-32E module and are therefore used here as an example.

HX711 pin	ESP32-DevKitC pin	Note
VCC	3V3	Keeps the DT signal at 3.3 V
GND	GND	Common ground
DT (data)	GPIO 16	Weight data to ESP32
SCK (clock)	GPIO 17	Clock from ESP32

Table 6. HX711 to ESP32 wiring (example pin choice).

5.3 Indicators and Power

Component	Connection
Red LED	GPIO 26, through 220 Ohm resistor, to LED anode; cathode to GND
Green LED	GPIO 27, through 220 Ohm resistor, to LED anode; cathode to GND
Buzzer	GPIO 25 to 1 kOhm to transistor base; buzzer positive to 5 V (VIN), buzzer negative to collector; emitter to GND
ESP32 data link	Micro-USB cable to laptop (serial data), or Wi-Fi to the laptop or broker
ESP32 power	Laptop USB port, or 5 V power bank through the micro-USB port
USB camera	USB cable directly to laptop

Table 7. Indicator, data and power connections.

The ESP32 board should be powered from only one supply at a time [4]. When the ESP32 is connected to the laptop by USB, the laptop supplies the power and the power bank is not needed; the power bank is used when the board communicates by Wi-Fi.

5.4 Load Cell Mounting

A bar load cell is mounted with one end fixed to the base and the other end free to move slightly under load. The 150 x 150 mm platform is fixed to the free end. Spacers are placed between the cell and both the base and the platform so that the cell body can flex without touching either. Items should be placed near the centre of the platform, because off-centre loads change the reading.

6. Communication, Software and Calibration

6.1 Communication Options

In Stage 1 the ESP32 sends each stable weight reading to the laptop as a short text message over USB serial. This needs no network and is the simplest to debug. In later stages the same message is sent over Wi-Fi using MQTT, as planned in the project report. An example message is:

{"cart_id": "01", "weight_g": 182, "delta_g": 182, "stable": true}

6.2 Software Tools

Tool	Use
Arduino IDE or PlatformIO with ESP32 core	Programs the ESP32
HX711 Arduino library (for example, the library by bogde)	Reads the HX711 and applies tare and scale
Python 3 with pyserial	Receives weight data from the ESP32 on the laptop
Ultralytics YOLOv8 and OpenCV	Camera input and product detection
paho-mqtt	Wi-Fi messaging in later stages

Table 8. Software tools.

6.3 Calibration Procedure

1. With the platform empty, record the raw HX711 value as the offset (tare).
2. Place a reference mass of known weight and record the new raw value.
3. Calculate the scale factor as (raw value - offset) divided by the known weight in grams.
4. Check the result with two or three other known masses.
5. Repeat the procedure whenever the mounting or platform is changed.

7. Practical Considerations

The HX711 produces 10 samples per second at its default rate, so a reading is available about every 100 ms [1]. The weight should be accepted only after several consecutive readings agree, because the platform needs a short time to settle after an item is placed. Averaging a few samples reduces noise. Load cell wires should be kept short and away from the USB and power wires. The cell has a rated capacity of 10 kg and is most accurate at about half of its rating [5], so a lower-capacity cell is preferable if the products to be weighed are light. Temperature changes and vibration will move the reading, which is why the camera result is used together with the weight.

8. Conclusion

The Stage 1 IoT chain consists of a Logitech C270 USB camera on the laptop, a TAL220 10 kg load cell read by an HX711 amplifier, and an ESP32-DevKitC-32E that processes the weight, drives the buzzer and LEDs and reports to the laptop. With the HX711 powered at 3.3 V, resistors on the LEDs, a transistor on the buzzer and the load cell correctly mounted, these parts are electrically compatible and the connections in Section 5 can be built as described. The corrections in Table 4 should be applied to the blueprint before the hardware is assembled.

References

[1] Avia Semiconductor, HX711 24-Bit Analog-to-Digital Converter (ADC) for Weigh Scales, datasheet. https://manualmachine.com/avia/hx711/28020542-datasheet/

[2] Components101, HX711 24-Bit Analog to Digital Converter. https://components101.com/ics/hx711-24-bit-analog-digital-converter-adc

[3] Grid Connect, ESP32-DevKitC-32E Development Kit with ESP32-WROOM-32E. https://www.gridconnect.com/products/esp32-devkitc-32e-wroom-module-development-kit

[4] Kobee, ESP32-DevKitC V4 Pinout. https://kobee.com.au/pinouts/esp32-devkitc-v4

[5] SparkFun, Load Cell - 10kg, Straight Bar (TAL220), SEN-13329. https://www.sparkfun.com/load-cell-10kg-straight-bar-tal220.html; Core Electronics, 10kg Beam Load Cell. https://core-electronics.com.au/makerverse-10kg-load-cell.html

[6] B&H; Photo Video, Logitech C270 HD Webcam. https://bhphotovideo.com/c/product/825671-REG/Logitech_960_000694_C270_HD_Webcam_Black.html